SheetShrink: Efficient Spreadsheet Cleanup

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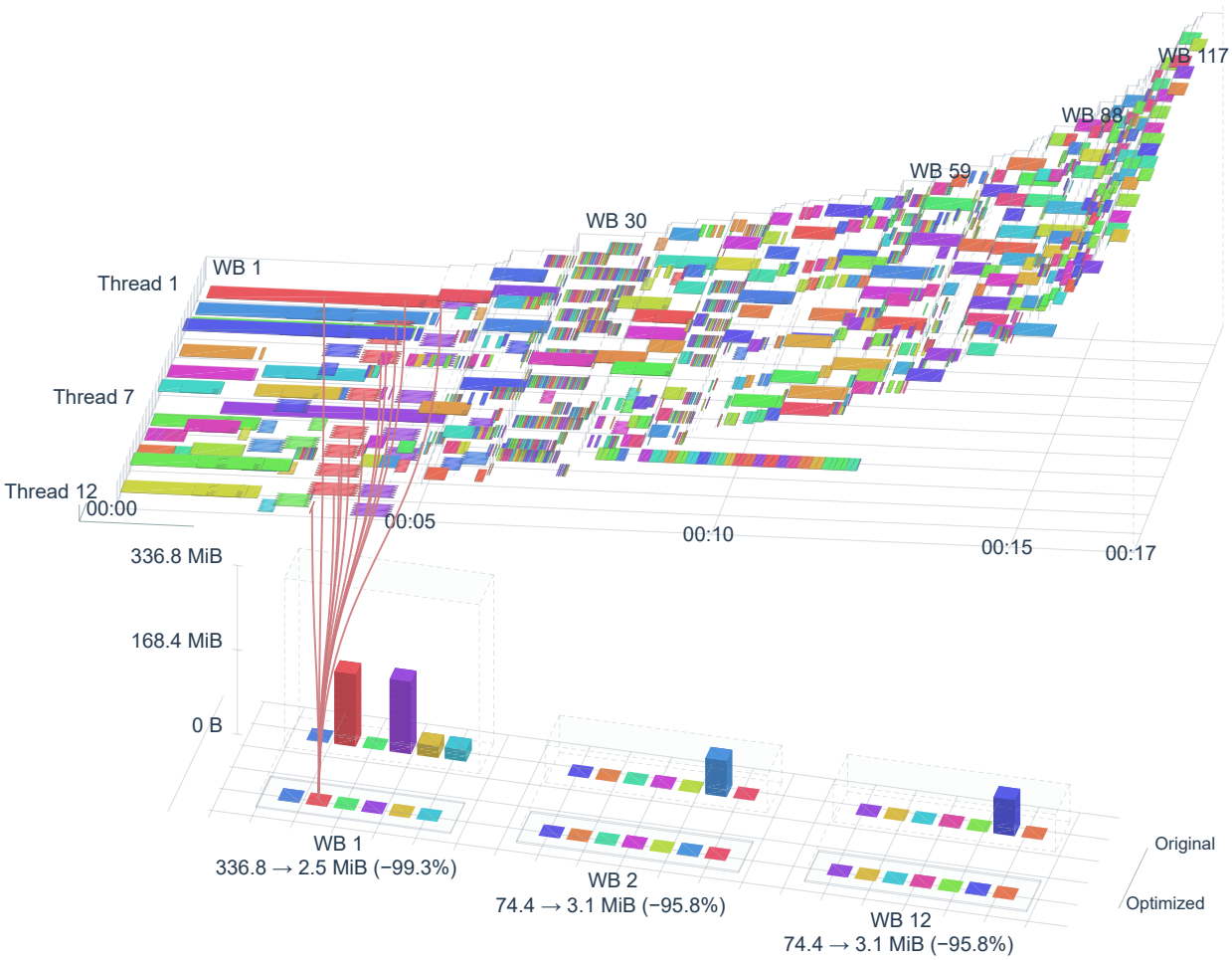


Fig. 1. Recorded development run on 144 Enron workbooks selected for large absolute XLSX reductions, using 12 workers. Work intervals follow workbook admission order (WB denotes workbook, numbered by admission). Thread labels identify worker lanes. The lower view compares the three largest uncompressed packages on one linear byte scale. Curves connect worksheet S2 in WB 1 to its work items. Observed wall time was 17.077 s.

***Abstract*—Spreadsheet packages can contain large populations of empty or redundant cell records that increase storage and ingestion costs. Cleanup must distinguish this structural redundancy from small formatting or dependency witnesses that preserve required workbook behavior. We present SheetShrink, a byte-oriented OOXML cleanup system that rewrites stored XLSX packages before opening them in Excel. It combines workbook-context analysis, local rewrite decisions, compact formatting representatives, and bounded parallel execution. We define its preservation contract and separate intentional policy choices from defects. This working manuscript reports bounded implementation checks and an ongoing native-rendering validation workflow. Controlled end-to-end performance results and final corpus coverage remain to be established.**

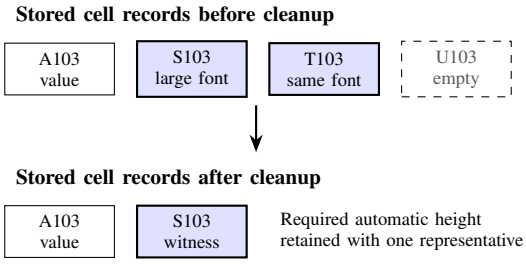


Fig. 2. Illustrative local cleanup. Numeric content and one required font witness remain. Redundant empty records are omitted while cell coordinates remain fixed. The preservation decision also depends on workbook context.

I. Introduction

Spreadsheet files are both data containers and executable visual documents. A user can store a compact table in a workbook whose worksheet parts contain large numbers of otherwise empty cells. Such records can increase the work required to decompress and interpret the package. Removing every empty-looking record is unsafe. An empty cell can preserve a border, override an inherited fill, or influence automatic row height through its font. A whitespace value can prevent text from extending into its neighbour. Cleanup therefore requires a model of the effects that stored structures have on the document.

For an XLSX input, SheetShrink makes these decisions from the stored package before loading the workbook in Excel. Its cleanup stage has no native rendering or calculated row-height observations. An Excel-hosted method can inspect the loaded layout and evaluate candidate edits against native observations. Our method instead retains compact source structures whose required effects follow from package analysis and the preservation rules. Native Excel is used later to evaluate the output. Legacy XLS conversion is a separately measured preparation stage.

Specialized spreadsheet parsing has demonstrated the importance of decompression, parsing, memory use and parallel execution for data loading [1]. Our problem adds a transformation obligation. The output remains an editable XLSX workbook and must retain required data, dependencies and visible presentation under a stated contract. File size alone is an inadequate measure of success. A smaller file that changes an important row height has failed that obligation.

Figure 2 illustrates the compact preservation approach. SheetShrink combines byte-oriented worksheet processing with compact preservation of necessary effects. It reuses existing styles and retains representative cells where those representatives preserve layout. Its production cleanup policy keeps whitespace blockers and applies dependency guards to eligible hidden constant tails. It leaves style and shared-string table compaction to a separate Excel save operation. This boundary keeps cleanup independent of the installed Excel application and makes its decisions inspectable in the package.

The performance objective includes the complete user operation. For a sufficiently bloated workbook, we seek to make cleanup followed by opening the cleaned file faster than directly opening its original. This objective must be measured, including the cases where cleanup adds overhead. A single-thread comparison separates improvements in work performed from gains due to extra processing resources. Parallel corpus experiments then examine throughput and memory.

The paper develops the preservation model, the small-footprint mechanisms that implement it, and an evaluation combining structural checks with native visual observations. The evaluation distinguishes completed observations, unsupported inputs and unresolved differences. Historical Python implementations provide context for the engineering progression. Their different cleanup rules require separate output and correctness accounting in any timing comparison.

II. Preservation and Comparison Contract

A. Observable spreadsheet behavior

A spreadsheet package contains more than the populated table that a user sees. Individual cell records can carry formatting, protection, references, or values that affect other cells. Conversely, the presence of a stored record does not establish that it contributes to the intended presentation. SheetShrink targets production cleanup of such packages while retaining the structures needed by its preservation contract. Aggressive historical configurations are separate test artifacts and are excluded from the production subject of this paper.

The relevant visual area includes cells with meaningful values or formulas, empty cells with effective visible fills or borders, and graphical objects. Formatting inherited from rows or columns can contribute to this area. Conditional formatting and text extending into adjacent cells can also affect what is visible. The union of the original and transformed relevant areas is required when checking removed content. Layout changes in otherwise empty trailing rows or columns are permitted by the contract. Changes that move or alter relevant content remain subject to validation.

B. Information available before native loading

The cleanup decision uses stored values, formulas, styles, references and layout attributes. Native effective row heights and candidate before/after images are available only in the separate validation workflow. Stored custom heights can be retained directly. For automatically sized rows, a font-bearing blank can still require a representative even when its payload is empty.

An Excel-hosted alternative could measure a row, remove candidates and write the measured height as an explicit custom height. SpreadsheetML provides a *customHeight* attribute for that policy [2]. Applying this technique requires a trusted height for the chosen renderer and a contract for replacing automatic sizing with a fixed height. The evaluated cleanup derives its compact preservation decisions from package context.

C. Removal decisions and compact preservation

A record can be eligible for removal only after accounting for its required contribution. An empty cell may still influence automatic row height through its font. A whitespace-only value may stop text from extending from an adjacent cell. Referenced hidden content can influence displayed values and charts. These cases require different reasoning from a test for an absent payload.

The preservation strategy retains a small representative when that representative can preserve a required effect. Row-font sentinels and guarded promotion of an existing style into a row format are examples. Their applicability depends on inherited formatting, gaps, merges, protection, and the available style records. An unresolved case retains the required source representation. The objective is to remove eligible bloat throughout the package while keeping the preservation footprint small.

The production policy admits separately guarded hidden constant tails and retains ordinary nonempty hidden tables, including unreferenced values. Tail admission depends on dependency and layout checks [3]. Absence of a reference alone does not authorize general hidden-data removal. A future Excel-hosted cleanup stage could measure native layout before replacing additional sentinels or whitespace blockers. Such a stage is outside the implementation evaluated here.

D. Validation and attribution boundaries

Our fast visual validation compares independently rebuilt original and SheetShrink states using the same installed Excel environment. Ordinary-save and rebuild controls retain information for attributing differences introduced by Excel. For XLS sources, conversion precedes the XLSX cleanup subject and has its own failure accounting. A file requiring repair or inaccessible protected content receives an explicit unsuccessful outcome.

Targeted visual equality is an observation over the selected regions. We evaluate the selection mechanism separately with native reference captures and mutations. Unequal pixels are retained for investigation. Repeated rendering controls, volatile dependencies, and capture geometry help classify their cause. The results section must report the completed coverage and unresolved findings at its fixed evaluation cutoff.

III. Method

A. Package-level analysis

Figure 3 summarizes the logical stages. Each line denotes a guarded operation whose preservation conditions are detailed below.

An XLSX file contains related ZIP members. SheetShrink first checks the supported serialization and reads the metadata needed to interpret worksheet decisions. The production parser targets unprefixed Transitional SpreadsheetML in its normal default namespace. Explicit prefixes on core cell structures, unsupported encodings and ambiguous representations can exclude an input. Exclusion retains a reason and does not establish that the workbook lacks redundant data.

Require: Workbook package W, cleanup policy P
Ensure: Cleaned package or explicit unsuccessful outcome
1: Check supported serialization of W
2: Resolve styles, strings, layout and references
3: **for all** worksheets in W **do**
4: Classify content and effective formatting
5: Identify protected references and layout anchors
6: Select guarded removals under P
7: Retain compact font and geometry witnesses
8: Rewrite admitted records and retain other effects
9: **end for**
10: Reconcile names and calculation-chain changes
11: Assemble output with unaffected package members
12: Publish only a successfully completed package

Fig. 3. SheetShrink's logical cleanup stages. Guards are resolved before removal. Worksheet scheduling and chunking preserve these decisions.

The analysis resolves cell styles, shared strings, merged regions and dependency sources before applying the relevant removal rules. Worksheet cells are processed as byte ranges. This avoids constructing a document object model for every cell. It still requires correct recognition of attributes, hierarchy and relevant namespace boundaries. Unsupported syntax follows the documented fallback or eligibility path. Byte-oriented execution is an implementation strategy and does not itself prove semantic preservation.

B. Classify effects before removing records

A value, formula or inline-string payload is content. Numeric zero and shared formula followers require explicit treatment. For a candidate empty cell, the cleaner considers explicit formatting together with its row and column context. An apparently default style can cancel a visible inherited format. Removing it would expose that format, so the effective effect matters.

A font can affect automatic row height without producing visible text. The cleaner therefore retains font representatives in its production mode. Where guards permit, an existing row style can replace repeated per-cell representations. This promotion must preserve other inherited components, gaps and merged cells. Dominance decisions include otherwise unsuitable styles, and ties have a fixed ordering. Ignoring an unsuitable majority would change the algorithm rather than merely accelerate it.

Whitespace cleanup is disabled in the production cleanup policy. A neighbour's text overflow and row sizing can depend on such values. Hidden constant-tail cleanup is separately guarded by references and layout conditions. Ordinary nonempty hidden tables remain outside that admitted tail policy. Protection and dependency decisions take precedence over a requested deletion.

C. Preservation structures at empty coordinates

The retained regression cases distinguish empty payloads from absent visual effects [3]. A default-format cell can mask a filled row or column. A font-only cell can determine automatic row height, including when a protection attribute accompanies the font. Merged text can require a local font representative even when no separate apply-font flag is present.

One font witness preserves row layout

A2 has Arial 10. Empty B2 and C2 have Arial 18.

Source

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="1"/>
- <c r="C2" s="1"/>
</row>
```

Guarded result

```
<row r="2">
 <c r="A2">...</c>
 <c r="B2" s="1"/>
</row>
```

Original control **23.25 pt**

Font witness control			
10 pt text			
Next visible row			

One witness retained **23.25 pt**

Font witness control			
10 pt text			
Next visible row			

Both witnesses removed **12.75 pt**

Font witness control			
10 pt text			
Next visible row			

Native Excel vector export after full rebuild. Removing the repeated witness is safe here. Removing both moves the next visible row.

Fig. 4. A reduced font-witness control. Red marks the removed duplicate. The source and one-witness native vector exports match after a full rebuild in Excel 16.0 build 20326. Removing both empty Arial 18 cells reduces row 2 from 23.25 to 12.75 points and moves the following row. XML is abbreviated. These are constructed native print controls with gridlines, separate from the corpus regression.

These cases require resolving inheritance before classifying removable cells. Figure 4 pairs a reduced XML example with a native Excel control. Retaining one font witness preserves the observed row height. Removing both changes the position of later visible content. The control is constructed independently of the corpus case and isolates this local effect.

Preservation can also depend on several coordinates together. Center-across- selection formatting requires the contiguous blank span following its content cell. Keeping only the last blank loses the intended span. Row-level borders can require widths at empty column boundaries. A chart's linked number format can depend on the first, empty series cell. The guards therefore preserve specific layout and dependency anchors beyond the nonempty-cell envelope. Figure 5 shows a retained Enron counterexample. The smaller Check Performance artifact exposes a different inherited fill. This observation concerns the pictured rebuilt region, with the historical producer identities retained alongside the captures.

The hidden constant-tail rule admits only a dense horizontal sequence of at least 4,096 identical signed-integer cells, hidden and ending at column XFD, subject to reference and layout guards. Gaps, formulas, text and vertical runs remain outside this rule. The threshold identifies an admitted structural case; it does not establish that arbitrary repeated data are unintended.

D. Guarded row representation

Promotion replaces repeated cell formatting with an existing row style only after checking the row and its gaps. The row must be visible, unprotected and without an existing row format. The current implementation additionally requires both a stored height and an enabled `customHeight` flag. An automatic row can change height after promotion even when its cells originally share the proposed font. A stored `ht` value by itself does not authorize promotion. The implementation checks these parsed attributes before the promotion scan, so the additional guard requires no native layout measurement [4].

The dominant empty-cell style must occur at least twice. The dense route requires at least 1,024 occurrences. A bounded-gap route admits at most 64 missing columns and restores required original formats only when the inheritance checks permit. Sparse promotion requires equivalent inheritance at every absent position and excludes merged sparse rows. Figure 6 shows the compact full-width representation. Style-count ties select the smaller existing style index. These thresholds choose a compact representation within the preservation conditions. They do not establish that arbitrary repeated values are redundant.

Automatic rows retain the ordinary representative rules. A column font can contribute to their height even when its cells contain no text. Representative selection must account for the inherited font identities that would disappear with deleted cells. An explicit default-style blank may also be the only barrier to a visible column fill or border. Retaining one required font witness does not authorize deleting a different cell that masks such inherited formatting.

E. Fixed height and local font witnesses

A fixed row height removes one reason to retain an empty font-bearing cell, but it does not eliminate all font effects. In a row containing text, the font of an empty cell can affect the text baseline even when the outer row height remains unchanged. The regression behind Figure 7 uses a nine-point fixed row in a university statistics workbook. Its retained blank is a local baseline witness. An empty fixed-height row has no

An empty gray cell masks an inherited pink fill

Column B uses XF 51 (pink). Cell B31 uses XF 98 (gray).

Original

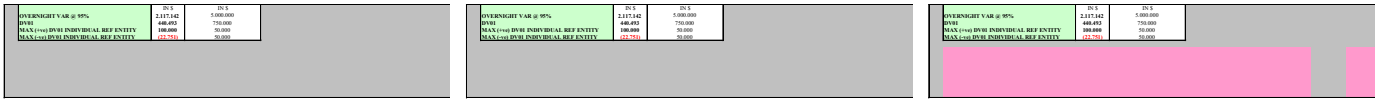
```
<c r="B31" s="98"/>
```

SheetShrink Default

```
<c r="B31" s="98"/>
```

Excel Check Performance

```
<!-- B31 removed -->
```



PL-Consolidated!A25:H37, vector export after full rebuild. Original and Default match pixel-for-pixel here. Removing the empty gray cells exposes the inherited pink background.

Fig. 5. Empty cells can preserve a visible fill. In Enron case `ccde4294bab9094d7d73`, an explicit gray B31 cell masks a pink column format. The retained Original and SheetShrink Default vector exports match in `PL-Consolidated!A25:H37` after full rebuild of the retained rebuild/Save controls. Check Performance removes the mask and exposes pink. The source excerpt is shortened. The print-renderer equality observation covers this region.

such text baseline, so a font-only blank can be removed when the other guards permit. This distinction prevents a height-only predicate from erasing a text-layout constraint.

A representative elsewhere in the row is insufficient when column inheritance creates a local gap. For example, two blank cells masking a column font can both be necessary even if a third cell already represents another font. The decision compares the effective styles at those coordinates. It does not infer local coverage from a row-wide set of font identifiers alone.

F. Promotion and complete inheritance coverage

Sparse promotion can reuse an existing font-equivalent style without moving unrelated rich cell formatting onto the row. Its gap checks therefore differ from simply counting repeated styles. Dense uniform-fill promotion also checks the content cells. Moving a fill onto the row would otherwise recolor a content cell whose original fill differs. Merged cells retain their required records. Figure 8 illustrates the admitted case with matching content fill and retained merged coordinates.

The count used to select a dominant style includes styles that subsequently fail the promotion guards. Filtering those styles before counting would allow an eligible minority to win and silently change the decision. A fixed tie-break on existing style indices makes selection deterministic. Unknown style indices, unsupported cell metadata and ambiguous attributes reject promotion and leave the ordinary cleanup path responsible for the row.

Redundant tail fills require complete column coverage. A white row fill can be redundant when every column through XFD already supplies the same fill. A gap of even one column breaks that argument. Column-tail coalescing likewise checks coverage and compatible effective formatting. Width differences beyond the retained content envelope can be irrelevant, while widths supporting visible row borders remain layout constraints. These are separate checks rather than a general rule that empty columns have no effect.

G. Protection beyond the content envelope

Worksheet protection changes the meaning of an otherwise empty coordinate. An unlocked or hidden-formula format can define an intended interaction region. The cleanup bounds distinguish the occupied data extent from a visually empty tail. Figure 9 shows an empty protected coordinate inside the retained extent and a removable counterpart outside it. Keeping every protection-bearing blank would preserve large accidental tails. Removing every such blank would discard the interaction boundary inside the sheet.

H. Deferred decisions and dependency scope

Reference-sensitive deletion occurs after candidate discovery. Otherwise a formula on a later worksheet could refer to a value already removed. Admitted hidden tails are stored as row spans, filtered against the protected references, and rewritten through the existing cleanup path. An intersecting known reference retains the entire span. This avoids materializing each candidate coordinate in a separate set and requires no additional traversal to rediscover the cells.

Reference protection includes supported formulas, names, chart and object formulas, and pivot dependencies. Shared-formula references move with relative axes while absolute axes remain fixed. The implementation can protect enclosing rectangles without expanding every follower. Conflicting anchors, unresolved scope, dynamic references and unsupported syntax prevent a deletion proof. Functions such as `INDIRECT` and `OFFSET` therefore require conservative handling [5].

Figure 10 makes the shared-formula distinction explicit. The relative row or column expands with the follower range, while a fixed axis remains fixed. Sheet scope is equally important. An unqualified reference on another sheet does not protect the same coordinates on the candidate sheet. Chart source

08 Move repeated formatting into the row

Source

```
  <row r="1" ht="9"
   customHeight="1"
  >
   <c r="A1" s="3"><v>7</v></c>
-  <c r="B1" s="2"/>
-  <!-- repeated through XFD1 -->
-  <c r="XFD1" s="2"/>
  </row>
```

Guarded result

```
  <row r="1" ht="9"
   customHeight="1"
+  s="2" customFormat="1"
  >
   <c r="A1" s="3"><v>7</v></c>
  </row>
```

Reuse an existing XF. Require a visible fixed-height, unprotected row and proved inheritance. Dense route needs at least 1,024 occurrences.

Unsafe shortcut. A newly inherited fill, border, protection or font at a gap can change the workbook.

Abbreviated XML. Style IDs are local.

Fig. 6. Guarded dense-row promotion reuses an existing XF and keeps the value cell. Red marks removed lines and green marks added lines. The illustration omits the repeated middle cells. The full-width regression removes 16,383 empty cell records under the fixed-height and inheritance guards.

31 Fixed height still needs a font witness

Source

```
  <row r="134" ht="9"
   customHeight="1">
   <c r="A134" s="7"><v>7896</v></c>
   <c r="L134" s="9"/>
-  <c r="M134" s="9"/>
  </row>
```

Guarded result

```
  <row r="134" ht="9"
   customHeight="1">
   <c r="A134" s="7"><v>7896</v></c>
   <c r="L134" s="9"/>
  </row>
```

Keep one explicit default-font witness in this text row, even though the height is fixed. Style 9 also has an explicit no-fill component.

Unsafe shortcut. Removing both blanks changes the text baseline. A fixed height alone does not prove identical text positioning.

Abbreviated XML. Style IDs are local.

Fig. 7. A fixed height does not eliminate a text-baseline witness. The retained empty cell preserves the font context of the fixed-height statistics row.

ranges and pivot source or output regions are resolved in their own scopes. Unresolved structured references and dynamic addressing prevent a safe negative reference result.

Candidate deletion and format deletion are separate operations. A diagnostic whitespace candidate can lose its value while retaining a fill that masks an inherited format. Figure 11 shows this phase-level test. Whitespace removal remains disabled in Default. The example documents the rewrite contract for an admitted candidate rather than broadening production eligibility.

After deferred deletion, affected rows are reconsidered because their earlier tail anchors may be stale. Row headers and column boundaries can still carry visible effects. In particular, an explicit borderless row can suppress column borders. A visible empty row with a stored height can determine the vertical extent of those borders. Preserving the row header is sufficient in these cases; the cleaner need not generate a full row of explicit cells.

The second cleanup of affected rows is consequential. An earlier candidate may have served as the last content anchor. Once it disappears, blanks retained solely because of that anchor can become removable. Reusing the stale bound would retain a tail that the final row no longer needs. The deferred regression checks this transition separately from initial candidate admission.

A trailing run of empty hidden rows can retain its final boundary while dropping intermediate records. The same rewrite is unsafe before later visible content, where row removal would alter placement. This policy preserves the admitted layout boundary and does not promise to preserve every exterior blank-grid observation.

I. Package metadata and failure boundaries

Workbook metadata cleanup has separate predicates. Built-in name aliases are matched by name and local sheet scope when a corresponding prefixed name exists. The current predicate does not establish equality of the two formula bodies. This remains a specific limitation of that metadata policy. Calculation-chain cleanup resolves removed cells through workbook relationships and retains unresolved mappings. The cleaner does not compute replacement formula caches.

J. Execution and output

The implementation distributes worksheet work and can subdivide large work. Scheduling and chunk size must preserve cleanup decisions. Row-aligned chunk boundaries retain the context needed by row-local decisions. Attribute scanning also respects quoted values. Text resembling `r="Z99"` inside

33 Uniform fill promotion keeps merge cells

Source

```
 <row r="1" ht="20.25"
  customHeight="1"
 >
  <c r="A1" s="2">...</c>
  <c r="C1" s="1"/>
  <c r="D1" s="1"/>
-  <!-- other cells also use s=1 -->
 </row>
 <mergeCell ref="C1:D1"/>
```

Guarded result

```
 <row r="1" ht="20.25"
  customHeight="1"
+  s="1" customFormat="1"
 >
  <c r="A1" s="2">...</c>
  <c r="C1" s="1"/>
  <c r="D1" s="1"/>
 </row>
 <mergeCell ref="C1:D1"/>
```

The full row has the same applied fill, including its content cells. Reuse XF 1 while preserving the merge's required explicit cells.

Unsafe shortcut. A different content-cell fill, visible border, border override or protection hazard prevents this promotion.

Abbreviated XML. Style IDs are local.

Fig. 8. Uniform-fill promotion checks content-cell fills and retains the merged coordinates. The red omission line represents the removed repeated cells.

35 Protection-only retention has a boundary

Source

```
 <row r="1"><c r="B1" s="7"/></row>
 <row r="2"><c r="C2"><v>1</v></c></row>
- <row r="3"><c r="D3" s="7"/></row>
```

Guarded result

```
 <row r="1"><c r="B1" s="7"/></row>
 <row r="2"><c r="C2"><v>1</v></c></row>
```

Default KeepInDataRange retains the protection-only B1 cell inside the later-established visible extent. D3 has no independent visual or dependency role.

Unsafe shortcut. Removing all protection-only cells loses in-range metadata. Keeping all of them can anchor a large irrelevant tail.

Abbreviated XML. Style IDs are local.

Fig. 9. Protection-sensitive empty cells are interpreted relative to the retained data extent. The outside-tail counterpart can be removed under the remaining guards.

a metadata attribute must not be mistaken for the cell coordinate. These parser invariants are independent of whether the corresponding cell is ultimately removable. Workbook-level concurrency also determines how many decompression buffers, analysis contexts and pending outputs coexist. Consequently, throughput experiments include mixed workbook sizes and single-sheet workbooks, alongside large multi-sheet inputs.

The writer copies unaffected package members and can reuse compressed entries where supported. It does not generally renumber style or shared-string tables. The output writer must preserve input bytes and existing outputs on a rejected operation. A partially processed package is not published as a successful cleanup. These invariants are checked separately from compressed file size.

K. Native validation

The companion DeltaSieve manuscript describes the selection and capture methodology [6]. Its working PDF is included in the supplementary material at related-manuscripts/deltasieve.pdf. We use native Excel rendering to investigate selected potentially affected regions. Original and cleaned copies are rebuilt in the same renderer environment. A separate validation methodology identifies value, effective-style, layout and dependency candidates. Its selection scope and failure accounting are part of the reported evidence. Independent reference samples and injected changes are required to evaluate that selector. Equality of its selected images alone cannot establish equality of every possible spreadsheet observation.

The companion figure atlas, figures/xml-methods/xml-cleanup-atlas.pdf, provides 49 abbreviated source/result examples, negative cases and a pinned source index. Its code is drawn at 8.2 points for a 252-point column. XML wrappers, irrelevant attributes and repeated payloads are omitted. Style indices are local to each example. Schematic effects are labeled separately from native Excel captures.

The coverage index accounts for all ten declared cleanup causes. This includes the disabled self-closing-cell preprocessing path and the currently inactive row-attribute cleanup helper. Their presence in an enumeration does not imply an active Default reduction. The atlas groups the 49 examples into seven rule families and labels internal candidate fixtures and optional policies. This coverage is an audit of the principal implemented decisions, with focused regression evidence. It is not an exhaustive proof over combinations of Excel features.

IV. Evaluation

The evaluation asks three questions. RQ1 examines compressed package reduction and the remaining preservation footprint. RQ2 measures cleanup time, memory and the combined cleanup-plus-opening operation. RQ3 examines struc-

40 Shared formulas preserve absolute axes

Source

```
<c r="B10"><f t="shared" si="7"
 ref="B10:D12">
Inputs!$E5+Inputs!F$6+Inputs!$G$7
</f></c>
<c r="D12"><f t="shared" si="7"/></c>
```

=

Retained by the guard

```
<c r="B10"><f t="shared" si="7"
 ref="B10:D12">
Inputs!$E5+Inputs!F$6+Inputs!$G$7
</f></c>
<c r="D12"><f t="shared" si="7"/></c>
```

Resolve the Inputs sheet and protect E5:E7, F6:H6 and G7 as the anchor moves over B10:D12. Fixed axes do not move.

Unsafe shortcut. Expanding both axes or binding the references to the formula's own sheet gives the wrong protected cells.

Abbreviated XML. Style IDs are local.

Fig. 10. Shared-formula followers move relative axes while retaining absolute axes. The protected source envelope follows those distinct rules.

44 Delete a value, keep its visible format

Diagnostic whitespace cleanup. Default keeps whitespace.

Source

```
 <row r="4">
  <c r="W4" s="91"><v>118</v></c>
-  <c r="X4" s="37" t="s">
-   <v>15801</v>
-  </c>
  <c r="Y4" s="91"><v>7</v></c>
 </row>
```

Guarded result

```
 <row r="4">
  <c r="W4" s="91"><v>118</v></c>
+  <c r="X4" s="37" t="s"/>
  <c r="Y4" s="91"><v>7</v></c>
 </row>
```

An already admitted, unprotected whitespace value is stripped. Re-cleaning retains the now-empty XF-37 cell because its fill is still visible.

Unsafe shortcut. Deleting the entire cell would conflate value removal with formatting removal. A font witness or dependency can prevent admission earlier.

Abbreviated XML. Style IDs are local.

Fig. 11. An admitted diagnostic whitespace candidate loses its payload while retaining a required format. Default keeps whitespace values.

tural preservation and observed native presentation under the stated validation scope.

A. Corpus identity and outcome accounting

The evaluation treats Data.gov, Destatis and Enron as distinct populations. Archive checksums, member names and duplicate occurrences identify each selected corpus version. Legacy XLS conversion is reported separately from XLSX cleanup. A historical Enron development selection contains 20,872 inputs, with 20,503 prepared artifacts and 369 unsuccessful preparation outcomes at its recorded preparation checkpoint. Its selected visual scope contains 16,799 workbooks. This run ended after the worker-pool investigation. These counts describe that development selection and do not establish completed visual coverage. The replacement EDRM Enron v2 XML handover contains 34,065 distinct content-addressed candidates totaling 8.470 decimal GB of uncompressed payload. The imported selection unit is a distinct candidate SHA-256. Candidate transport was checked against the supplied manifests, and the import retained the original candidate bytes. This externally generated extraction includes legacy Excel and other OLE candidates. Native format recognition, successful XLSX conversion and visual eligibility remain separate evaluation stages. Candidate count therefore describes the imported input frame, with workbook outcomes to be reported after processing. The upstream extraction provenance and local import identities are retained alongside the experiment.

The current native visual stress cohort selects the 49 whole workbooks with the largest absolute compressed-file byte reduction in each corpus, totaling 147 original identities. All processor baselines use those same identities. Earlier development observations outside this reduced cohort remain retained history. Availability of an optimized variant and completion of its visual comparison are reported separately.

B. Recovered chart-data workbooks

Recursive extraction also discovers Microsoft Graph OLE objects that Excel rejects as chart gallery files. An optional preparation stage embeds each object in a temporary PowerPoint presentation, requests native chart conversion, and extracts the generated XLSX data workbook. These derived chart-data workbooks form a separate population from the original Excel workbooks.

A subsequent diagnostic run retained slide exports from ten additional conversions after a three-second rendering delay. Inspection revealed outputs with large black regions and fragmented chart geometry, including the example in Figure 12. These observations motivate excluding Graph-derived workbooks from the main preservation evaluation pending source-to-output fidelity validation. The original objects remain avail-



Fig. 12. PowerPoint rendering after converting an extracted legacy Microsoft Graph object. Large black regions obscure fragmented chart geometry. The image is a cropped native slide export produced three seconds after conversion with PowerPoint 16.0. Source rendering fidelity remains unverified.

TABLE I
FROZEN XLSX SOURCE INVENTORIES. SIZES ARE DECIMAL GB OF LISTED WORKBOOK FILES AND DO NOT MEASURE COMPLETED NATIVE COVERAGE.

Corpus	XLSX members	XLSX total (GB)
Data.gov	9,117	25.273
Destatis	1,182	2.364

able for inspection. The cause of the observed rendering remains unresolved because a verified rendering of the source object has not yet been established.

An exploratory checkpoint on November 10, 2026 included all 969 successful recoveries available at that point, without random sampling. SheetShrink's default policy left 884 packages byte-identical and changed worksheet XML in 85 packages (8.8%). The cleanup counters recorded 368 empty cells without content, 153 empty hidden cells, and 48 trailing layout-only rows removed. Aggregate worksheet XML decreased from 3,515,260 to 3,499,682 bytes. Compressed package size decreased from 8,831,892 to 8,671,550 bytes (1.82%), which also reflects package recompression. Cell-value and formula payloads matched before and after cleanup in all 969 files, and all changed package parts were worksheets. Native visual equivalence has not been evaluated for this subset.

This checkpoint establishes that native-generated chart-data workbooks can retain removable structure. It does not estimate its prevalence among failed, unconverted or remaining Graph objects. The modest aggregate reduction motivates keeping this recovery stage optional. Per-input identities, binary identity, configuration and measurements are retained in `evidence/graph-recovery-pilot-20261110.json` and the accompanying CSV.

The two frozen XLSX inventories in Table I come from the versioned source profiles. Each listed archive member has an individual SHA-256 identity. Their source-frame sizes are separate from the eligible, completed and independently referenced subsets.

A dated evaluation cutoff will report compared workbooks, differences, failures, unsupported representations and unfinished work. Repeated attempts have their own counts. An unchanged package can bypass visual work under an exact input-identity check, but a failed capture cannot contribute an equality observation. The full recursive Enron XML-derived corpus is a separately identified population.

C. Sampling and reference controls

The random evaluation sample is selected from a frozen eligible frame with a recorded seed before inspecting visual findings. Strata use corpus identity, package size and structural features available before validation. Low-reduction and unchanged files remain represented. A separately labelled development stress set contains large and historically problematic workbooks.

Feasible sampled files receive full native reference captures over the relevant visual area using a capture plan independent of the candidate selector. A full-reference timeout remains in the denominator. It is not replaced silently by an easier workbook. Mutations with known visible effects cover cell values, styles, geometry, conditional dependencies and objects. Mutations that Excel normalizes away are reported separately.

D. Performance and comparison subjects

The comparison records each method's input representation, Excel dependency, use of native layout observations, removal policy and resulting workbook properties. The first-opening objective measures cleanup followed by loading the resulting workbook against direct loading of the source. An Excel-hosted cleaner includes its required source opening in the corresponding workflow. Cleanup, opening, saving and any required reopening have separate phase timings under a common readiness endpoint. Repeated-use experiments report preparation cost and subsequent opening cost separately.

Production cleanup time covers package analysis and output construction. Native preservation-validation cost has its own accounting. A corpus workbook supplies an observed source state. Establishing preservation requires structural checks and independently prepared native controls. Synthetic stress cases vary eligible redundant records, inheritance masks, font witnesses, merged spans and protected references, with fixed-height and automatic-height rows distinguished. Report removed records and bytes together with retained witnesses, exclusions and preservation outcomes under each cleanup configuration.

For each input, report the original and cleaned compressed sizes and their ratio. Report distributions and corpus totals, including inputs with no reduction. Opening time uses a documented readiness signal and fresh owned Excel processes. Single-thread cleanup, cleanup with parallelism and corpus throughput are separate experiments. RAM measurements

TABLE II
SELECTED DESTATIS STRESS PANEL. ORIGINAL SIZE IS DECIMAL MB. OTHER COLUMNS GIVE PACKAGE REDUCTION (%). SS DENOTES SHEETSHRINK, NX DENOTES NXPOWERLITE, CP DENOTES CHECK PERFORMANCE, AND XL DENOTES XLSTYLESTOOL.

Workbook	Original	SS	NX	CP	XL
Finance	89.37	94.85	97.09	96.85	0.01
Traffic	37.21	92.99	65.83	94.04	0.02
Migration	16.93	90.67	30.64	89.27	0.02
Electricity	1.43	89.86	91.92	63.96	0.11
Doctors	15.24	78.22	79.52	82.00	0.02
Time use	3.84	81.38	83.14	81.66	0.32

specify the sampling interval and whether they refer to startup, steady state or peak use.

Microsoft Check Performance is an optional cleanup comparison subject. Its Excel save-only control is necessary to distinguish optimization from serialization and recalculation effects. Its availability does not guarantee that all selected files can be processed. Historical DOM and regular-expression Python implementations have different cleanup decisions. A timing comparison must therefore include output differences and preservation outcomes for each version.

E. Present verified evidence and remaining measurements

Focused implementation verification includes 292 passing Core library tests and one ignored benchmark at the recorded development revision. A separate exploratory comparison of an attribute-scan proposal found identical decompressed outputs on seven hash-matched review inputs, over two alternating-order repetitions. The two large cases showed essentially unchanged wall time. The proposal was not adopted. These observations establish a bounded engineering result and do not estimate corpus-wide speedup or visual correctness.

Evaluation freeze pending. Populate the corpus reduction, end-to-end opening, resource and independent-reference result tables from frozen manifests. Development-run counters are not used to claim an absence of regressions.

F. Six-file cleanup stress panel

A separate stress panel selects six Destatis workbooks from a 100-file historical-reduction screening, including the Migration workbook's hidden zero tail. Each cleaner receives a separate hash-verified original copy. SheetShrink uses revision `a7eef1a`. The comparison subjects are NXPowerLite Desktop 11.0.1 with its Balanced profile, native Excel Check Performance in Excel 16.0.20527.20150, and XLStylesTool UWP 3.0.1.0 with optional style normalization, name unhiding and wholesale external-name removal disabled. All 18 external cleanup operations produced retained XLSX outputs with valid ZIP CRCs. Table II reports compressed package sizes. The artifact records full member names, input/output hashes and exact settings.

The Migration workbook contains 343,833 numeric zero constants in hidden columns of worksheet `Tabelle 3.4`. NXPowerLite, Check Performance and XLStylesTool retain all of them. SheetShrink removes these constants under its unreferenced-hidden-content policy, reducing that worksheet from 361,086 to 787 stored cells while retaining the hidden-column interval. This package observation distinguishes the cleanup footprints even when package reduction alone appears similar.

Native rendering also exposes formatting witnesses. In the Doctors workbook, NXPowerLite removes empty 11-point-font cells Q36:AB36 from worksheet `10.2 T1`. The stored automatic-height row remains, but rebuilt Excel measures row 36 at 12 points instead of the original 15 points. An independent `A1:AC41` repeat retains identical original images across fresh sessions. Original against original has zero differing pixels. NXPowerLite and Check Performance differ in 47,383 and 34,762 pixels, respectively, inside the common image rectangle. SheetShrink has zero differing common pixels and a one-pixel white bottom-extent difference. This diagnostic was selected after observing the initial finding.

For Migration, a separate original-derived reference plan covers 48 rectangles over the visible value/formula envelopes of its 36 worksheets. NXPowerLite and Check Performance change overlapping pixels in two and one rectangles, respectively. SheetShrink and XLStylesTool preserve common pixels in all 48, with white extent differences accounted for separately. The repeated original image identities agree across all four subjects. Formatting-only exterior tails, overflow beyond supplied rectangles and outlying object effects remain outside this reference scope. A targeted Traffic attempt exceeded available memory, and targeted Migration attempts encountered full-grid and idle-progress limits. Their failures remain recorded alongside bounded reference follow-ups. These observations describe the selected stress panel. Corpus prevalence, exhaustive rendering coverage and matched execution costs remain separate measurements.

G. Automatic-height regression and correction

The Traffic workbook supplies a concrete counterexample to promotion based only on repeated font formatting [4]. On worksheet `3.8_(1)`, row 105 originally inherits a 12.75-point height without an explicit height or row style. The earlier cleanup promotes its dense empty-cell format into the row. Excel then measures 9 points, moving the following footer. The independently supplied `A1:U110` reference rectangle contains 3,408 changed pixels in the common image area, and its candidate image is five pixels shorter. This is a native presentation failure with retained input, producer and image identities.

Requiring an explicit custom height before promotion restores the original 12.75-point row. Fresh rebuilt original and corrected images both measure 1,358 by 1,327 pixels and are byte-identical. The corrected workbook occupies 2,654,536 bytes against a 37,207,883-byte original. The retained development cleanup took 13.28 seconds with statistics enabled. This single observation does not estimate guard overhead. The source change checks two existing parsed attributes before promotion and retains fixed-height promotion coverage in the regression suite. This correction postdates the six-file panel's

pinned a7eef1a cleaner and therefore does not retroactively replace its outputs.

A separate current-cleaner Migration repeat produced 1,804,772 bytes from the 16,926,818-byte source. Under DeltaSieve policy v12, 37 image pairs had no pixel differences. A former 12.75-point discrepancy in a zero-width range disappeared when its empty lower context row was excluded. Separate effective-height observations on two other sheets remain in the report. Consequently this repeat supports the recorded image scope and the corrected exclusion, with those remaining observations explicitly retained [7].

H. Check Performance preparation checkpoint

A completed Destatis preparation campaign attempted all 1,182 source inputs and retained 1,016 prepared outputs and 166 failures. Its raw failure reasons contain 165 opening timeouts and one automation failure when setting Excel's Interactive property. These are preparation outcomes. Visual validation of the outputs and matched baseline cost measurements remain open.

Recovery checks of the same 166 frozen failure inputs used an owned opening-dialog monitor. The first three inputs completed in 13.213 seconds and the remaining 163 in 442.761 seconds, each batch with two workers. The checks identified Name Conflict dialogs for all 165 former opening timeouts. The Interactive-property failure recurred. A separate known-positive input produced its prepared output in 8.114 seconds. Source-identity reconciliation accounts for every original failure, while conflicting-name contents remain unresolved. Worker count changed during the first campaign, so its aggregate elapsed time is observational. Original and recovery receipts remain separately versioned alongside the 1,016 successful preparation artifacts.

A later eleven-input retry on the selected stress panel again produced only automation failures. One Enron workbook nevertheless completed Check Performance in the author's manual session. Its source SHA-256 matched the frozen input, and the saved output reduced serialized cells from 1,315,602 to 1,094,514. The automated pane had reported "Something went wrong" for the same input, including in a fresh Excel process. This discrepancy remains an unresolved workflow difference. It cannot establish that Check Performance is unable to clean the workbook. The manual artifact has separate provenance and pending visual validation.

I. Validation cost of redundant background definitions

The Destatis workbook anbau-gartenbaugewaechse-2030224169005.xlsx contains repeated column-width records over a uniform full-width background. Rebuilt original and SheetShrink outputs contain 128,912 and 2,197 column definitions, respectively. A corrected DeltaSieve background-envelope guard reduced the native selection from 16,101 to 165 regions. All 165 image pairs were identical in the fresh comparison. This observation supports the recorded targeted scope. It does not replace the remaining corpus validation or establish exhaustive visual preservation.

a) Explicit no-fill font representatives.: The Destatis workbook pruefungen-hochschulen-2110420217005.xlsx exposed a reproducible text-baseline regression on sheet TAB-04. An original-versus-original reference control was pixel-identical, whereas the transformed output repeatedly changed 25,640 pixels despite equal capture geometry. Restoring 87 removed blank cells with an explicit default font and no-fill metadata recovered the original image hash. The correction allows this metadata combination to supply the existing per-row font representative, including fixed-height rows with content. A fresh output from the corrected cleaner matched the original over the independent A1:L916 reference area. The corpus's conflicting legacy print-name aliases were explicitly normalized for opening, with separate hashes and preserved name definitions. The core regression suite passed 297 tests with one fixture-dependent test skipped.

b) Automatic height without an explicit font-application flag.: In kulturfinanzbericht-gesamtmaterial-1023003229005.xlsx, removing the blank cell Impressum!B16 reduced rebuilt row 16 from 15 to 12.75 points. Its XF references Arial MT 12 without an explicit applyFont, whereas the text cell uses Arial 10. The stored ht="15" has no customHeight and does not freeze automatic sizing. The correction admits this non-default font to the existing per-row representative policy and still removes redundant representatives. A separate native control rebuilt both complete workbooks and captured the preselected Impressum!A1:J43 area. The 832 by 743 images were byte-identical with equal geometry. This local control completed in 13.07 seconds. A subsequent complete targeted comparison under capture policy version 15 completed all 350 selected workbook regions without a pixel, geometry or rendering difference. Historical findings remain retained with their original planning policy.

J. Repeated process measurements with historical Inquire observations

An additional experiment covers the 46 previously missing Inquire cases from the frozen top-49 selection. It includes 17 Data.gov, 28 Enron and one Destatis workbook. This missing-result subset is not a representative corpus sample. The recorded machine is an HP EliteBook 645 G11 with an AMD Ryzen 5 PRO 7535U, six physical cores, 12 logical processors, 32 GiB RAM and local NVMe storage. It runs 64-bit Windows 11 Enterprise 25H2. Excel Professional Plus 2024 has file version 16.0.17932.21056 and Inquire version 16.0.16626.20076.

SheetShrink commit 75e85035 was built in release mode for x86_64-pc-windows-gnu and executed natively on Windows. The experiment uses 12 Rayon threads, 12 writer workers and one active workbook. Each workbook receives one warmup and seven measured process invocations. Four

additional measurements are taken when the initial interquartile range exceeds 10% of the median. One case meets this rule, giving 46 warmups and 326 measured trials. File order is deterministically shuffled between rounds. Inputs and outputs use a warm filesystem cache without forced cache flushing. Process timing includes startup and file I/O and excludes subsequent hashing, verification and database writes. The controller samples process resources at approximately 20 ms.

All 46 SheetShrink cases completed. Per-file process-time medians range from 0.399 to 28.218 seconds. Independently recomputing medians from the retained trial table reproduces the reported values. Each retained output's uncompressed member fingerprint agrees with its repeated trials and with the corresponding output from the current cleaner build. Visual validation remains a separate requirement.

The historical Inquire observations were not repeated. They include 24 complete successes, three partial cleanups with warnings, 18 failed attempts and one explicitly unsupported outcome. An Inquire workflow time includes worker startup, copying and hashing, Excel opening, cleanup, saving, validation and process termination. For the 24 complete successes, the median across files of single Inquire workflow time divided by SheetShrink process-time median is 13.38, with a range of 3.72 to 31.67. These are descriptive ratios with different measurement boundaries and repetition counts. They are not an isolated algorithm-speedup estimate. Partial or failed Inquire attempts receive no ratio. Inquire's approximately 250-ms sampled memory maxima also differ from SheetShrink's OS-reported lifetime peaks.

Fifteen Inquire failures were dialog-appearance timeouts. Fourteen of these inputs store a chart sheet as their active tab, and the supplied harness does not activate a worksheet before invoking cleanup. These observations leave an invocation hypothesis unresolved. The timeouts do not establish that Inquire cannot clean the workbooks. Two other attempts failed during opening, and one unexpectedly opened read-only. The original attempts and errors remain part of the evidence.

V. Related Work

A. Workbook bloat and cleanup tools

Microsoft Check Performance removes cells associated with unnecessary formatting and other stored information from a loaded workbook [8]. Its documentation warns that optimization can remove pixel-art cells. Inquire's Clean Excess Cell Formatting command removes excess cells beyond the last nonblank cell and can change conditional-formatting coverage [9]. These documented policies motivate comparing both reduction and preservation outcomes under explicit contracts.

Package-level cleanup also predates SheetShrink. XLStylesTool's developer describes an Open-XML-based cleaner for excessive and corrupted styles that operates independently of Excel's object model [10]. NXPowerLite Desktop 10 lists excess XLSX cell-format cleanup among its features [11]. Its engine description discusses formatted blank cells as a source of large worksheet parts [12]. These are relevant comparison subjects. Their available descriptions do not establish the guarded removal rules, compact row-font witnesses or native preservation outcomes examined here. Those properties require version-specific evidence. SheetShrink's positioning therefore includes its effect-aware rules, retained representatives and measured costs alongside its execution before native loading.

B. Spreadsheet loading and specialized parsing

SheetReader studies specialized spreadsheet ingestion and the coupling of parsing, decompression and parallel execution [1]. It motivates evaluating loading cost and memory. SheetShrink produces an editable workbook whose required layout and dependencies must survive package rewriting. Its evaluation measures cleanup together with subsequent use of the output.

Mison combines structural indexing with speculative access to queried JSON fields [13]. Simdjson uses SIMD operations in a validating JSON parser [14]. These systems motivate efficient processing of semi-structured representations. SheetShrink's document-rewriting obligation additionally accounts for inheritance, references and native layout effects.

C. Data transformation and spreadsheet refactoring

Harris and Gulwani synthesize table transformations from example input/output pairs [15]. Potter's Wheel integrates interactive transformation and discrepancy detection [16]. HoloClean repairs data through probabilistic inference combining constraints and statistical evidence [17]. These systems infer or specify intended changes to data. SheetShrink's target is removal of admitted package redundancy under a preservation contract. Their data-repair accuracy measures therefore require a different ground truth from our reduction, cost and preservation measurements.

Hermans et al. detect formula smells and suggest refactorings to improve spreadsheet maintainability [18]. This is relevant to reasoning about workbook structure and behavioral obligations. Formula readability and refactoring objectives do not establish that empty cell records are removable while retaining native layout. Execution outside Excel alone likewise leaves the required effect analysis to be specified.

D. Validation and dependency analysis

Spreadsheet slicing integrates dependency reasoning with testing and debugging [19]. Dependencies also constrain removal when retained formulas or graphical objects refer to candidate content. Workbook comparison tools provide observations about cell and format differences [20]. Our native validation protocol uses DeltaSieve's region selection and capture workflow [6], with independent references and mutation controls to evaluate selected coverage. These validation mechanisms provide evidence about cleanup outcomes after the package transformation.

VI. Discussion and Limitations

The preservation contract is narrower than all possible spreadsheet behavior. Rendering depends on the installed

Excel version, fonts and calculation state. A selected-region observation has limited scope. Unsupported XML and unsuccessful opening or saving remain visible outcomes. The independent reference sample estimates behavior on its eligible population and cannot establish a universal absence of regressions.

Large reductions can coexist with small but important preservation structures. A future Excel-hosted stage could replace font sentinels with measured custom row heights and validate the replacement immediately. Such a stage has a native application dependency and is outside the cleanup implementation examined here. Whitespace removal likewise requires a separate justification of overflow and layout effects. It is not assumed safe merely because a displayed value is blank.

The historical implementations explain the progression from diagnosis through regular-expression cleanup to specialized byte-oriented execution. They are useful engineering artifacts with distinct output behavior. Their age or slower runtime does not establish that the current algorithm is correct.

Regression cases can support a reusable methodological observation when their mechanism and control are retained. The Traffic example identifies a failure of automatic-height style promotion, while the Migration example identifies an exterior-context classification error in the validator. Their remedies operate at different stages. Preserving that distinction prevents a validator change from being reported as a cleaner improvement. Operational failures in the native comparison environment also affect achieved evaluation coverage and belong in the recorded outcomes even when cleanup itself completed quickly.

VII. Conclusion

SheetShrink addresses spreadsheet redundancy through effect-aware removal and compact preservation. Its implementation and evaluation distinguish compressed size, XML changes, execution cost and observed presentation. The completed manuscript must quantify those outcomes using the frozen experiments described above and state the remaining unsupported cases.

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